2. Find the area bounded by the lines $y-2x-4=0, y=1,\ y=3$ and the y-axis

3. Calculate the area bounded by the parabola $y^2 = 4ax$ and its latusrectum.

4. Find the area bounded by the line $y = x$, the x-axis and the ordinates $x = 1$, $x = 2$.

5. Using integration, find the area of the region bounded by the line $y-1=x, the\ x\ axis$ and the ordinates $x = -2$, $x = 3$.

6. Find the area of the region lying in the first quadrant bounded by the region $y=4x^2, x=0, y=0$ and $y = 4$.

7. Find the area bounded by the curve $y=x^2$ and the line $y = 4$

3.2 Application of Integration in Economics and Commerce.

Integration helps us to find out the total cost function and total revenue function from the marginal cost. It is possible to find out consumer's surplus and producer's surplus from the demand and supply function. Cost and revenue functions are calculated through indefinite integral.

We learnt already that the marginal function is obtained by differentiating the total cost function. Now we shall obtain the total cost function when marginal cost function is given, by integration.

3.2.1 Cost functions from marginal cost functions

If C is the cost of producing an output x, then marginal cost function $MC=\frac{dc}{dx}$. Using integration, as the reverse process of differentiation, we obtain,

Cost function $C=\int (MC)\,dx+k$

Where k is the constant of integration which is to be evaluated,

Average cost function $AC=\frac{C}{x},\ x\neq 0$

Example 3.9

The marginal cost function of manufacturing x shoes is $6+10x-6x^2$. The cost producing a pair of shoes is ₹12. Find the total and average cost function.

Solution:

Given,

Marginal cost $MC = 6+10x-6x^2$

$$C = \int MC\,dx + k$$
$$= \int \left(6+10x-6x^2\right) dx + k$$
$$= 6x+5x^2-2x^3+k \quad (1)$$

when $x = 2$, $C = 12$ (given)

$$12 = 12+20-16+k$$
$$\boxed{k=-4}$$
$$C = 6x+5x^2-2x^3-4$$

Average cost $= \frac{C}{x} = \frac{6x+5x^2-2x^3-4}{x}$

$$= 6+5x-2x^2-\frac{4}{x}$$

Example 3.10

A company has determined that the marginal cost function for a product of a particular commodity is given by $MC=125+10x-\frac{x^2}{9}$ where C rupees is the cost of producing x units of the commodity. If the fixed cost is ₹250 what is the cost of producing 15 units.

Solution:

$$MC = 125+10x-\frac{x^2}{9}$$
$$C = \int MC\,dx + k$$
$$= \int \left(125+10x-\frac{x^2}{9}\right) dx + k$$

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